

Exploring the Link Between Out-of-Distribution Detection and Conformal Prediction with Illustrations of Its Benefits

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Abstract

Research on Out-Of-Distribution (OOD) detection focuses mainly on building scores that efficiently distinguish OOD data from In Distribution (ID) data. On the other hand, Conformal Prediction (CP) uses non-conformity scores to construct prediction sets with probabilistic coverage guarantees. In other words, the former designs scores, while the latter designs probabilistic guarantees based on scores. In this paper, we study how these two fields can benefit each other, and we formalize several aspects of this connection. First, **for OOD detection**, we show that in standard OOD benchmark settings, evaluation metrics can be affected by the validation dataset’s finite sample size. Extending the work of [Bates et al. \(2023\)](#), we define new *conformal AUROC* and *conformal FPR@TPR95* metrics, which are corrections that provide probabilistic guarantees on the variability of the FPR involved in these metrics with respect to the validation datasets. We show the effect of these corrections on two reference OOD and anomaly detection benchmarks, OpenOOD ([Yang et al., 2022](#)), and ADBench ([Han et al., 2022](#)). Second, **for CP**, we study the use of OOD scores as non-conformity scores and show that they can improve the efficiency of the prediction sets obtained with CP in several settings.

Keywords: Conformal prediction, Out-of-distribution detection, classification, conformal p-values.

1. Introduction

Even though current Machine Learning (ML) and Deep Learning (DL) models are able to perform several complex tasks that previously only human beings could, we are still a step away from their widespread adoption in safety-critical applications. Indeed, it is difficult to certify a ML component, mainly due to the poor control of the circumstances that may provoke such a ML component to fail. Out-of-Distribution (OOD) detection tries to tackle this problem by identifying data that differs significantly from the data used to train the model at runtime. Besides being recognized as an essential step in the certification of ML systems by multiple certification authorities (see, e.g., Sections 5.3 and 8.4 of ([Balduzzi et al.,](#)

2021) or Section 5.1 of (EASA and Daedalean, 2024)), OOD detection is a very active branch in machine learning research.

Current OOD detection strategies rely on constructing an OOD score s , a function that assigns a scalar to each input example. This score discriminates between in-distribution (ID) data and OOD data by assigning lower scores to the former and higher scores to the latter.

When OOD detection is used in a machine learning pipeline to identify examples that differ from the data the model has been trained on, there is a natural qualitative interpretation of OOD detection in terms of model uncertainty. For instance, an example with a low OOD score should be one for which the model can predict with low uncertainty, while an example with a high OOD score should be linked to a highly uncertain prediction. Conformal Prediction (CP) is a family of post-hoc methods for Uncertainty Quantification and Uncertainty Representation (Caprio et al., 2024), that work as wrappers over machine learning models, transforming point predictions into prediction sets with rigorous probabilistic guarantees based on so-called nonconformity scores. The user pre-specifies a risk level α , and the constructed prediction set is guaranteed to contain the ground truth value with a probability of at least $1 - \alpha$.

Since CP is a way of providing rigorous uncertainty quantification guarantees built upon scores, it is natural to apply it to the scores used in OOD detection. Conversely, CP relies on nonconformity scores, most of which are close to existing OOD scores. **This paper studies this connection and highlights several ways in which CP and OOD may benefit from each other.**

First, we start from the work of Bates et al. (2023) on outlier detection. Bates et al. (2023) cast the outlier detection problem into the statistical framework of hypothesis testing. They show that the p-values, built with a calibration dataset, are provably marginally valid but depend on the choice of the calibration dataset, and so does the False Positive Rate (FPR) derived from these p-values. We show that this effect can introduce fluctuations in the approximation of subsequent OOD metrics like AUROC and FPR@TPR95, thus overestimating or underestimating the true metrics. In a certification process, we are mainly interested in guaranteeing that our estimations are conservative with high probability, at the expense of losing some approximation precision (EASA, 2023). On the other hand, when the focus lies in performance and OOD score benchmarking, we would prefer to check if the most competitive baselines should legitimately be considered as such, despite the fluctuations. To achieve that, we design new confidence interval bounds-like metrics for AUROC and FPR@TPR95 based on CP theory: AUROC^- and FPR@TPR95^- that are conservative estimates and AUROC^+ and FPR@TPR95^+ that are permissive estimates, with a user-chosen probability $1 - \delta$. We show the effect of these new metrics on two large reference benchmarks, the OOD benchmark OpenOOD (Yang et al., 2022), and the anomaly detection benchmark (Han et al., 2022).

Second, we show that not only can CP contribute to OOD detection, but research in OOD detection can also help CP. Indeed, CP has traditionally focused on constructing prediction sets from nonconformity scores. Still, the scores used are usually simple functions of the softmax scores for classification tasks or classical distances in Euclidean space for regression tasks. Here, we draw inspiration from the OOD detection literature to build more involved nonconformity scores and compare their performance to the traditional nonconformity scores of CP. For the task of classification, we build prediction sets based on multiple different OOD

scores and find that some of them, notably Mahalanobis (Leys et al., 2018) or KNN (Sun et al., 2022), are good candidates as nonconformity scores.

Our two contributions are two sides of a single observation: **OOD detection is about designing scores, while CP is about interpreting scores to produce guarantees.** The first contribution instantiates the CP→OOD direction (using CP theory to interpret and correct the scores that OOD evaluation relies on), and the second the OOD→CP direction (feeding OOD scores into CP as nonconformity scores). Making this score-to-guarantee bridge explicit is what enables cross-fertilization between the two fields. In summary, our contributions are the following:

- We adapt conformal corrections to standard OOD evaluation metrics and study their effect on OOD evaluation and benchmarking.
- We evaluate several OOD scores as nonconformity scores for conformal classification and identify settings in which they improve prediction-set efficiency.

2. Background

Out-of-Distribution Detection Given n examples, $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ sampled from a probability distribution $\mathcal{P}_{id}$ on a space $\mathcal{X}$, and a new data point $\mathbf{x}_{n+1}$, the task of Out-of-Distribution (OOD) detection consists in assessing if $\mathbf{x}_{n+1}$ was sampled from $\mathcal{P}_{id}$ - in which case it is considered In-Distribution (ID) - or not - thus considered OOD.

The most common procedure for OOD detection is to construct a score $s : \mathcal{X} \rightarrow \mathbb{R}$ and a threshold τ such that:

$$\begin{cases} \mathbf{x}_{n+1} \text{ is declared OOD if } s(\mathbf{x}_{n+1}) \geq \tau \\ \mathbf{x}_{n+1} \text{ is declared ID if } s(\mathbf{x}_{n+1}) < \tau \end{cases} \quad (1)$$

We call s an OOD score.

Task-based OOD This is the most common approach in the literature regarding OOD detection for neural networks. It also encompasses Open-Set Recognition. Let's consider that $\mathbf{x}_i$ can be assigned a label y_i so that we can construct a dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ defining some supervised deep learning task. In that case, $\mathcal{P}_{id} := \mathcal{P}_{train}$. Task-based OOD uses representations built by the neural network f throughout its training to design s . Many sophisticated methods follow this approach (Yang et al., 2021). A simple example is to take the negative maximum of the output of f (after the softmax) (Hendrycks and Gimpel, 2018) as an OOD score ($s(\mathbf{x}_{n+1}) = -\max(f(\mathbf{x}_{n+1}))$) where $\max(\mathbf{x})$ is the highest component of the vector $\mathbf{x}$. Another simple idea is to find the distance to the nearest neighbor in some intermediate layer of f (Sun et al., 2022).

Task-agnostic OOD This approach encompasses One-Class Classification and Anomaly or Outlier Detection. Let's consider a dataset $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ in a fully unsupervised way. There is no notion of labels, so we have to approximate $\mathcal{P}_{id}$ somehow or some related quantities from scratch. Examples are GANs or VAEs with s defined as reconstruction error. See (Yang et al., 2021) for a thorough review.

Conformal Prediction Few Machine Learning and Deep Learning models provide a notion of uncertainty related to their predictions. Even the models trained for classification tasks providing softmax outputs, which can be interpreted as the probabilities for the input belonging to the different classes, are usually ill-calibrated and overconfident, making the softmax output an incorrect proxy of the true uncertainty of the prediction (Pearce et al., 2021). Conformal Prediction (CP) (Vovk et al., 2005; Angelopoulos and Bates, 2022) is a series of post-processing uncertainty quantification techniques that are model-agnostic and provide finite-sample guarantees on the model predictions. One of the simplest CP techniques, the split CP, works as a wrapper on a trained model f . It requires a calibration dataset $\{(\mathbf{x}_{n+1}, y_{n+1}), \dots, (\mathbf{x}_{n+n_{\text{cal}}}, y_{n+n_{\text{cal}}})\}$ independent of the training data, and a risk (or error rate) α that the user can tolerate. Based on so-called nonconformity scores computed on the calibration dataset, it builds a prediction set $C_\alpha(\mathbf{x}_{n+n_{\text{cal}}+1})$ for a new test sample $\mathbf{x}_{n+n_{\text{cal}}+1}$ with the following finite sample guarantee

$$\mathbb{P}(y_{n+n_{\text{cal}}+1} \in C_\alpha(\mathbf{x}_{n+n_{\text{cal}}+1})) \geq 1 - \alpha. \quad (2)$$

To obtain the guarantee in equation (2), the only assumption required is that the calibration and test data form an exchangeable sequence (a condition weaker than, and therefore automatically satisfied by independence and identical distribution) (Shafer and Vovk, 2008) and that they are independent of the training data. It is essential to know that the guarantee in equation (2) is marginal, i.e. holds in average over both the calibration dataset and the test sample choice. As we emphasize below, finite calibration sets induce fluctuations that should be accounted for in practice.

3. Related Works

In this work, we study the use of Conformal Prediction as a statistical framework for interpreting OOD scores. This idea of casting OOD in a statistical framework has already been attempted in different settings.

Selective Inference and Testing Selective Inference works on top of a ML predictor by using an additional decision function to decide for each example whether the original model’s prediction should be considered. A score equivalent to an OOD score is used to define this decision function. Several approaches exist, for instance, through building a statistical test (Haroush et al., 2022) or by training a neural network with an appropriate loss (Geifman and El-Yaniv, 2017, 2019). In our setting, the framework of Conformal Prediction is particularly suitable because it applies to scores in a post-processing manner, does not require assumptions or modifications on the model, and yields finite-sample guarantees.

Conformal Prediction with Reject Option and Selective Prediction Closely related to selective inference is the conformal reject-option setting, which sits directly at the intersection of the two perspectives we study. Conformal Prediction naturally induces an abstention rule: a test point yielding a singleton prediction set can be predicted confidently, whereas an empty or a multi-label set signals that the model should abstain or reject. In this way, the structure and size of the CP sets act as a distribution-free uncertainty signal that plays a role analogous to an OOD score (Guan and Tibshirani, 2022). Building on this idea, Szabadváry et al. (2025) formalize binary classification with a reject option by accepting only

singleton conformal sets, and derive distribution-free guarantees on the resulting error rate; Garcia-Galindo et al. (2024) extend the reject-option framework to multi-class classification with accuracy and recall guarantees at a prescribed rejection rate, and Xu et al. (2025) study selective conformal risk control. These works decide *whether* to predict from the structure of the conformal set, a decision rule conceptually close to OOD detection and abstention. Our contributions are complementary and act on the score itself: we use CP theory to correct OOD *evaluation* metrics, and we use OOD scores as CP *nonconformity* scores. Combining the two, e.g. building reject-option classifiers whose nonconformity scores are OOD scores, is a natural direction that our results directly enable.

Conformal OOD and AD Conformal Prediction has been previously applied to Out-of-Distribution and Anomaly Detection. For instance, Liang et al. (2024) have proposed a method based on CP for OOD with labeled outliers, and Kaur et al. (2022) and Zhang et al. (2025) propose to use conformal p-values. CP is one of several frameworks that allow obtaining statistical guarantees for OOD detection. One of the first methods for Anomaly Detection was introduced by Vovk et al. (2003). Since then, several other methods have been proposed by Laxhammar and Falkman (2011); Laxhammar (2014); Balasubramanian et al. (2014), as well as more recently by Angelopoulos and Bates (2022); Guan and Tibshirani (2022), where the lengths of the prediction sets act as OOD scores, or Bashari et al. (2025) where the ID dataset is assumed to be contaminated by outliers. These works all use the standard CP setting, in which basic marginal guarantees are obtained. We go further on this approach by using CP as a probabilistic tool to refine the interpretation and, hence, the usefulness of any OOD score.

Finding Efficient Scores for Conformal Prediction We also study the use of OOD scores as non-conformity scores in CP. In this sense, OOD scores can be viewed as a broader family of candidate scores for CP, which opens an additional avenue for score design. Common ways to build prediction sets for classification, such as LAC (Sadinle et al., 2019) or APS (Romano et al., 2020) and RAPS (Angelopoulos et al., 2020) are based on the softmax output of classifiers. However, non-conformity scores also exist for other predictors (Vovk et al., 2005), for instance, based on nearest neighbor distance (Shafer and Vovk, 2008). As shown by Johansson et al. (2017), the choice of the nonconformity score has a major impact on the efficiency. Our contribution lies on giving a principled approach to build a nonconformity score from an existing OOD score.

4. OOD Scores Through the Lens of CP

Let us begin by describing the typical benchmark setup for evaluating an OOD score. First, an OOD detector is fit on $\mathcal{D}_{id}^{train} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. Then, the OOD score is evaluated on $\mathcal{D}_{id}^{val} = \{\mathbf{x}_{n+1}, \dots, \mathbf{x}_{n+n_{val}}\}$ and $\mathcal{D}_{ood} = \{\bar{\mathbf{x}}_1, \dots, \bar{\mathbf{x}}_{n_{val}}\}$, where $\mathcal{D}_{ood}$ is a dataset sampled from a different distribution $\mathcal{P}_{ood} \neq \mathcal{P}_{id}$ (typically, another dataset). We apply s to obtain $\{s(\bar{\mathbf{x}}_1), \dots, s(\bar{\mathbf{x}}_{n_{val}}), s(\mathbf{x}_{n+1}), \dots, s(\mathbf{x}_{n+n_{val}})\}$. Then, we assess the discriminative power of s by evaluating metrics depending on a threshold τ . By considering ID samples as negative and OOD as positive, we can compute:

- The Area Under the Receiver Operating Characteristic (AUROC): we compute the False Positive Rate (FPR) and the True Positive Rate (TPR) for $\tau_i = s(\mathbf{x}_{n+i})$,

$i \in \{1, \dots, n_{\text{val}}\}$, and compute the area under the curve with FPR as x-axis and TPR as y-axis.

- FPR@TPR95: since the TPR is the detection rate on OOD data, the threshold τ is set to the empirical 5%-quantile of the OOD scores $\{s(\bar{\mathbf{x}}_1), \dots, s(\bar{\mathbf{x}}_{n_{\text{val}}})\}$, so that a fraction 0.95 of the OOD samples exceed it (TPR= 0.95); FPR@TPR95 is then the FPR evaluated on the ID validation data at this τ . It can be generalized to FPR@TPR $_{\beta}$, for any $\beta \in (0, 1)$.

A crucial step in any of these metrics is to compute the FPR. The FPR and its empirical estimation $\widehat{\text{FPR}}(\tau)$ are defined as follows:

$$\text{FPR}(\tau) = \mathbb{P}_{\mathbf{x} \sim \mathcal{P}_{id}}(s(\mathbf{x}) \geq \tau), \quad \widehat{\text{FPR}}(\tau) = \frac{1}{n_{\text{val}}} \sum_{i=1, \dots, n_{\text{val}}} \mathbf{1}_{s(\mathbf{x}_i) \geq \tau}. \quad (3)$$

4.1. OOD detection and p-values

Let us now rewrite the problem of OOD detection using the framework of statistical hypothesis testing. This framework allows us to reason in terms of p-values, which have multiple benefits: they have a rigorous mathematical definition and probabilistic interpretation, they can be interpreted equivalently for any score, and used for comparison of different scores. Given a test example $\mathbf{x}_{\text{test}}$, we wish to test for $\mathbf{x}_{\text{test}} \sim \mathcal{P}_{id}$, i.e. we wish to test the null hypothesis $\mathcal{H}_0 : \mathbf{x}_{\text{test}} \sim \mathcal{P}_{id}$ against the alternate hypothesis $\mathcal{H}_1 : \mathbf{x}_{\text{test}} \not\sim \mathcal{P}_{id}$. The value $P_{\mathbf{x} \sim \mathcal{P}_{id}}(s(\mathbf{x}) \geq s(\mathbf{x}_{\text{test}}))$ is an exact p-value for the null hypothesis $\mathcal{H}_0$. Note that this p-value corresponds to $\text{FPR}(s(\mathbf{x}_{\text{test}}))$ as defined in equation (3). Hence, the values $\widehat{\text{FPR}}(\tau_1) = \widehat{\text{FPR}}(s(\mathbf{x}_{n+1})), \dots, \widehat{\text{FPR}}(\tau_p) = \widehat{\text{FPR}}(s(\mathbf{x}_{n+n_{\text{val}}}))$ used in every OOD detection benchmark to compute the AUROC and FPR@TPR $_{\beta}$ can be considered as approximate p-values. The relationship between the FPR and the p-values emphasizes the link between OOD detection evaluation and hypothesis testing.

4.2. Fluctuations of the p-values and Impact on the FPR

This is where the framework of Conformal Prediction comes into play. Since we do not have access to the distribution $\mathcal{P}_{id}$, we approximate the FPRs (so the p-values) by using the validation dataset $\mathcal{D}_{id}^{\text{val}}$, which allows using results from CP to improve the evaluation of the FPR. Note that $\mathcal{D}_{id}^{\text{val}}$ can be related to the calibration dataset used in CP.

In this part, we mainly explain the work of [Bates et al. \(2023\)](#) that emphasizes that the p-value, and therefore the FPR, fluctuates depending on $\mathcal{D}_{id}^{\text{val}}$. We illustrate this phenomenon in the context of OOD detection and adapt the corrections proposed in ([Bates et al., 2023](#)) to this field by defining new conformal AUROC and conformal FPR@TPR $_{\beta}$.

Note first that the FPR can also be defined using a threshold t applied to the estimated p-values $\hat{u}^{\text{marg}}(\mathbf{x})$ as:

$$\text{FPR}(t, \mathcal{D}_{id}^{\text{val}}) = \mathbb{P}_{\mathbf{x} \sim \mathcal{P}_{id}} \left(\hat{u}^{\text{marg}}(\mathbf{x}) \leq t \mid \mathcal{D}_{id}^{\text{val}} \right), \quad (4)$$

where $t \in [0, 1]$. The authors point out that due to the empirical estimation of the p-value with $\hat{u}^{\text{marg}}(\mathbf{x})$, the quantity $P_{\mathbf{x} \sim \mathcal{P}_{id}}(\hat{u}^{\text{marg}}(\mathbf{x}) \leq t \mid \mathcal{D}_{id}^{\text{val}})$ is a random variable that depends on $\mathcal{D}_{id}^{\text{val}}$.

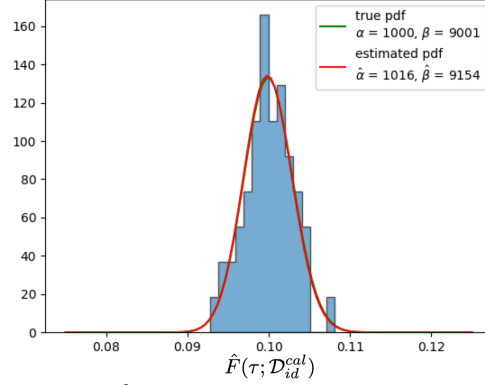


Figure 1: Histogram of $F(0.1; \mathcal{D}_{id}^{val})$ for different validation sets. The histogram is obtained by splitting the dataset *svhn_extra* into disjoint validation sets of 10000 points each, and approximating the value of F for each validation set by integrating over the remaining 521131 examples.

As a practical consequence, the FPR will fluctuate depending on which dataset $\mathcal{D}_{id}^{val}$ it is evaluated. The random variable $\text{FPR}(t, \mathcal{D}_{id}^{val})$ follows a distribution that is known: it is a Beta distribution that depends on the parameters n_{val} and t :

$$\text{FPR}(t, \mathcal{D}_{id}^{val}) \sim \text{Beta}(\ell, n_{\text{val}} + 1 - \ell), \quad (5)$$

where $\ell = \lfloor (n_{\text{val}} + 1)t \rfloor$ (cf. (Bates et al., 2023) or (Vovk, 2012) for a proof of the result).

4.2.1. ILLUSTRATION ON SVHN

To illustrate why this phenomenon matters in OOD detection, we leverage the fact that the SVHN dataset provides an additional set of 530000 *extra* test images. It allows the simulation of 53 draws of the random variable $F(t; \mathcal{D}_{id}^{val})$, by splitting the over 530000 examples in the *svhn_extra* dataset into 53 different folds of 10000 examples each. For each fold, the 10000 examples are used to constitute the calibration dataset $\mathcal{D}_{id}^{val}$, whereas the remaining over 520000 examples are used to approximate the computation of F , i.e., given a calibration dataset $\mathcal{D}_{id}^{val}$,

$$F(t; \mathcal{D}_{id}^{val}) \approx \hat{F}(t; \mathcal{D}_{id}^{val}) = \frac{1}{520000} \sum_{i=1 \dots 520000} \mathbf{1}_{\hat{u}^{\text{marg}}(\mathbf{x}_i) \leq t}. \quad (6)$$

Due to the large number of points used in the approximating sum, the 53 values obtained are faithful approximations of the random variables $F(t; \mathcal{D}_{id}^{val})$.

We perform this simulation with $t = 0.1$ and plot the 53 values into a histogram. Additionally, we fit a Beta distribution to the histogram using the *scikit-learn* library. These plots are found in figure 1. As we can see, the estimated parameters of the fitted beta distribution are very close to those predicted by the theoretical result of equation (5). If the value $\hat{u}^{\text{marg}}(\mathbf{x})$ were a true p-value, the value of $F(t; \mathcal{D}_{id}^{val})$ would be equal to t , but as we can see from the theoretical result and the experiment above, $F(t; \mathcal{D}_{id}^{val})$ is a random variable that fluctuates around its mean value t . This phenomenon can be detrimental to safety-critical applications, which are the applications of choice for OOD detection. Indeed, it may result

in underestimating the FPR, whereas we would like the FPR to be conservative. Conversely, it can also hinder performance evaluation of OOD scores by overestimating the FPR.

4.2.2. PROBABILISTIC GUARANTEES FOR P-VALUES AND THE FPR

To solve this problem, [Bates et al. \(2023\)](#) further corrects the marginal p-values, thus obtaining *calibration-conditional* p-values. Given a user-predefined risk level δ , the calibration-conditional p-values $\hat{u}^{cc-}$ will satisfy

$$\mathbb{P}\left(\mathbb{P}(\hat{u}^{cc-}(\mathbf{x}) \leq t \mid \mathcal{D}_{id}^{val}) \leq t, \forall t \in (0, 1)\right) \geq 1 - \delta, \quad (7)$$

where the probability inside is taken over $\mathbf{x} \sim \mathcal{P}_{id}$, and the probability outside over the choice of $\mathcal{D}_{id}^{val}$. Thus, with a probability of at least $1 - \delta$, we can be confident that our p-values will be **conservative**. This correction also applies to the FPR, which then satisfies the following:

$$\mathbb{P}\left[\text{FPR}(\tau) \leq \widehat{\text{FPR}}^-(\tau), \forall \tau \in \mathbb{R}\right] \geq 1 - \delta, \quad (8)$$

where $\widehat{\text{FPR}}^-(\tau)$ is a corrected version of the empirical $\widehat{\text{FPR}}(\tau)$. The corrected p-values and FPR, $\hat{u}^{cc-}(\mathbf{x})$ and $\widehat{\text{FPR}}^-(\tau)$, are obtained by applying a correction function h^- such that $\hat{u}^{cc-}(\mathbf{x}) = h^- \circ \hat{u}^{\text{marg}}(\mathbf{x})$ defined in [Appendix A](#).

If the focus of OOD score evaluation is performance or benchmarking, it is also useful to derive a lower bound on the FPR. The previous ideas extend naturally to this case. We therefore introduce a new correction for $\hat{u}^{\text{marg}}$ that we denote $\hat{u}^{cc+}$, which satisfies the property:

$$\mathbb{P}\left(\mathbb{P}(\hat{u}^{cc+}(\mathbf{x}) \leq t \mid \mathcal{D}_{id}^{val}) \geq t, \forall t \in (0, 1)\right) \geq 1 - \delta, \quad (9)$$

and likewise, we can define $\widehat{\text{FPR}}^+$ that satisfies

$$\mathbb{P}\left[\text{FPR}(\tau) \geq \widehat{\text{FPR}}^+(\tau), \forall \tau \in \mathbb{R}\right] \geq 1 - \delta, \quad (10)$$

thanks to the following proposition:

Proposition 1 *Let $0 \leq b_1 \leq b_2 \leq \dots \leq b_n \leq 1$ such that*

$$\mathbb{P}\left[U_{(1)} \leq b_1, \dots, U_{(n)} \leq b_n\right] \geq 1 - \delta.$$

Let $h^+ : \mathbb{R} \mapsto \mathbb{R}$ defined by

$$h^+(t) = 1 - b_{\lfloor (n+1)(1-t) \rfloor}, t \in [0, 1],$$

Suppose that the distribution of $s(\mathbf{x})$ is continuous. Then, $\hat{u}^{cc+}$ defined as $\hat{u}^{cc+}(\mathbf{x}) = h^+ \circ \hat{u}^{\text{marg}}(\mathbf{x})$ satisfies equation (9) and $\widehat{\text{FPR}}^+$ defined as $\widehat{\text{FPR}}^+(\tau) = h^+ \circ \widehat{\text{FPR}}(\tau)$ satisfies equation (10).

The notation $^+$ and $^-$ may not seem natural because $\widehat{\text{FPR}}^-$ is an upper bound and $\widehat{\text{FPR}}^+$ is a lower bound, but we use them to anticipate their effect on the AUROC, which will be inverted (see next section). In the following, we refer to the quantity $\widehat{\text{FPR}}^-(\tau) = h^- \circ \widehat{\text{FPR}}(\tau)$ (already defined in (Bates et al., 2023)) and $\widehat{\text{FPR}}^+(\tau) = h^+ \circ \widehat{\text{FPR}}(\tau)$ (introduced in this paper) as *conformal FPR* $^-$ and *conformal FPR* $^+$.

Four different correction functions h^- are proposed by Bates et al. (2023), the Simes, DKWM, Asymptotic and Monte Carlo corrections. The Simes, DKWM and Monte Carlo corrections all provide the finite sample guarantees of equation (7) and equation (8), while the Asymptotic correction provides only an asymptotic guarantee, that is, when the number of calibration points goes to infinity. Between the three corrections providing the finite sample guarantee, we find the Monte Carlo one to give tighter $\widehat{\text{FPR}}^-(\tau)$ and the DKWM one tighter $\widehat{\text{FPR}}^+(\tau)$ (please see Appendix C for more details on how the Simes and Monte Carlo corrections are defined.) The function h^- is trivially obtained from h^+ once the corrections are computed.

4.3. Conformal Metrics for OOD

Based on the previously defined conformal FPR^+ and FPR^- , we define *conformal AUROC* $^+$ and *conformal AUROC* $^-$ and *conformal FPR@TPR95* $^+$ and *conformal FPR@TPR95* $^-$. These quantities are obtained similarly to their classical versions, but using the conformal FPR^+ and FPR^- :

- *Conformal AUROC* $^+$ and *conformal AUROC* $^-$: we compute the *conformal FPR* $^+$ or *conformal FPR* $^-$ for $\tau_i = s(\mathbf{x}_{n+i})$, $i \in \{1, \dots, n_{\text{val}}\}$ and the True Positive Rate (TPR) for each of these values. We then compute the area under the curve with *conformal FPR* as x-axis and TPR as y-axis.
- *Conformal FPR@TPR95* $^+$ and *conformal FPR@TPR95* $^-$: We set τ to the empirical 5%-quantile of the OOD scores so that the True Positive Rate (TPR) is 0.95, and then compute the corresponding *conformal FPR* $^+$ or *conformal FPR* $^-$ on the ID validation data. It can be generalized to *FPR@TPR* β , for any $\beta \in (0, 1)$.

The computations are performed by considering the ID validation dataset as the calibration dataset. We would like to insist on the fact that conformal FPRs, AUROCs, and *FPR@TPR95*s are not necessarily better approximations of the real FPR, AUROC and *FPR@TPR95* values. Nonetheless, *FPR@TPR95* $^-$ and *AUROC* $^-$ are guaranteed to use conservative estimates of the FPR with a user-defined miscalibration tolerance δ , which is an essential property in many safety-critical applications or certification processes (Sellke et al., 2001). On the other hand, *FPR@TPR95* $^+$ and *AUROC* $^+$ provide a more permissive estimate that allows for evaluation and comparison of OOD scores that take this miscalibration effect into account. In short: the “ $-$ ” metrics (*AUROC* $^-$, *FPR@TPR95* $^-$) are the ones to report for conservative, safety-oriented evaluation, while the interval [*AUROC* $^-$, *AUROC* $^+$] (and likewise for *FPR@TPR95*) is the tool for benchmarking, as two scores whose intervals overlap cannot be distinguished at confidence $1 - \delta$.

Remark 2 (Which guarantee transfers to *AUROC* $^\pm$ and *FPR@TPR95* $^\pm$) *It is important to be precise about what the corrections do and do not guarantee. The corrections act*

only on the FPR (i.e. the ID side): Equations equation (8) and equation (10) state that, with probability at least $1 - \delta$, $FPR(\tau) \leq \widehat{FPR}^-(\tau)$ (resp. $FPR(\tau) \geq \widehat{FPR}^+(\tau)$) simultaneously for all τ , under continuity of the score distribution and exchangeability of the ID validation data. Because this bound is uniform in τ , it also holds at the data-dependent threshold used to compute $FPR@TPR95$, so $FPR@TPR95^-$ (resp. $FPR@TPR95^+$) bounds the FPR at that operating point with probability at least $1 - \delta$; note, however, that the operating point is fixed by the empirical $TPR = 0.95$ measured on the finite OOD sample, which itself carries sampling error that is not corrected. The TPR is never conformally corrected. Consequently, $AUROC^-$ and $AUROC^+$ should be read as AUROCs computed with conservative, resp. permissive, FPR estimates: they inherit the guarantee only along the FPR axis and are not certified distribution-free lower/upper bounds on the true AUROC.

Remark 3 (Conformal metrics do not require extra calibration data) *Computing the conformal FPRs only requires a correction to the estimated FPR. It does not require extra calibration data and applies to the FPRs computed with validation data. This is not like in CP, where we need a calibration dataset to find a threshold based on nonconformity scores obtained on calibration data, which is subsequently used to provide CP confidence intervals. Here, there are no confidence prediction intervals; we only use CP theory to obtain probabilistic guarantees for the FPR.*

4.4. Improved Benchmarks for OOD

AUROC and (to a lesser extent) $FPR@TPR95$ are two metrics that OOD and AD practitioners intensively use to benchmark and evaluate the performance of different OOD detection algorithms. However, as we saw in the previous sections, the evaluation can be sensitive to the choice of the ID validation dataset, which can be detrimental either to algorithms designed for safety-critical applications or for performance assessment. In this section, we reevaluate various OOD baselines included in the very furnished OpenOOD (Yang et al., 2022), and ADBench (Han et al., 2022) benchmarks, and illustrate the trade-off between performance and probabilistic guarantees. All our experiments can be easily carried out on a standard laptop CPU, once all the OOD scores are computed.

4.4.1. OPENOOD

OpenOOD (Yang et al., 2022) is an extensive benchmark for task-based OOD, i.e. for OOD methods that assess if some test data resembles some trained backbone’s training data. Usually, backbones trained on CIFAR-10, CIFAR-100, Imagenet200, and Imagenet are considered. In our case, we consider a ResNet18 trained on the first three datasets only since we are not evaluating a new baseline but only investigating new metrics for the benchmark. We evaluate the AUROC of several baselines with various OOD datasets gathered into two groups, Near OOD and Far OOD, following OpenOOD’s guidelines. We then compute the correction for the AUROC, with $\delta = 0.01$. The results are displayed in Table 1. We also run the benchmark for $\delta = 0.05$ and $FPR-95$, which we defer to Appendix E.

Two observations can be drawn from Table 1. *For safety concerns*, Table 1 shows that after the correction, the conformal $AUROC^-$ is lower than the classical AUROC, by often

	CIFAR-10		CIFAR-100		ImageNet-200	
OOD type	Near OOD	Far OOD	Near OOD	Far OOD	Near OOD	Far OOD
OpenMax (Bendale and Boulton, 2015)	87.2 ^{88.88} _{85.95}	89.53 ^{91.22} _{88.3}	76.66 ^{78.36} _{74.95}	79.12 ^{80.83} _{77.52}	80.4 ^{82.08} _{78.82}	90.41 ^{92.05} _{88.77}
MSP (Hendrycks and Gimpel, 2016)	87.68 ^{89.20} _{86.56}	91.0 ^{92.47} _{89.98}	80.42 ^{82.06} _{78.93}	77.58 ^{79.24} _{76.0}	83.3 ^{84.87} _{81.85}	90.2 ^{91.57} _{88.83}
TempScale (Guo et al., 2017)	87.65 ^{88.89} _{86.55}	91.27 ^{92.5} _{90.3}	80.98 ^{82.3} _{79.51}	78.51 ^{79.84} _{76.95}	83.66 ^{85.23} _{82.21}	90.91 ^{92.27} _{89.53}
ODIN (Liang et al., 2018)	80.25 ^{81.72} _{79.04}	87.21 ^{88.86} _{86.26}	79.8 ^{81.46} _{78.3}	79.44 ^{81.08} _{77.92}	80.32 ^{81.89} _{78.85}	91.89 ^{93.14} _{90.59}
MDS (Lee et al., 2018)	86.72 ^{88.28} _{85.49}	90.2 ^{91.72} _{89.09}	58.79 ^{60.49} _{56.85}	70.06 ^{71.72} _{68.31}	62.51 ^{64.21} _{60.68}	74.94 ^{76.58} _{73.09}
MDSEns (Lee et al., 2018)	60.46 ^{62.11} _{58.69}	74.07 ^{75.4} _{72.72}	45.98 ^{47.68} _{43.97}	66.03 ^{67.69} _{64.43}	54.58 ^{56.27} _{52.76}	70.08 ^{71.66} _{68.35}
Gram (Sastry and Oore, 2020)	52.63 ^{54.31} _{50.69}	69.74 ^{71.33} _{68.11}	50.69 ^{52.39} _{48.69}	73.97 ^{75.39} _{72.63}	68.36 ^{69.99} _{66.74}	70.94 ^{72.48} _{69.3}
EBO (Liu et al., 2020)	86.93 ^{88.32} _{85.9}	91.74 ^{93.0} _{90.9}	80.84 ^{82.49} _{79.36}	79.71 ^{81.38} _{78.19}	82.57 ^{84.15} _{81.1}	91.12 ^{92.49} _{89.71}
GradNorm (Huang et al., 2021)	53.77 ^{55.39} _{51.92}	58.55 ^{60.15} _{56.76}	69.73 ^{71.37} _{68.11}	68.82 ^{70.44} _{67.19}	73.33 ^{74.86} _{71.85}	85.29 ^{86.52} _{83.99}
ReAct (Sun et al., 2021)	86.47 ^{87.87} _{85.41}	91.02 ^{92.31} _{90.12}	80.7 ^{82.36} _{79.23}	79.84 ^{81.51} _{78.32}	80.48 ^{82.06} _{79.0}	93.1 ^{94.37} _{91.79}
MLS (Hendrycks et al., 2022)	86.86 ^{88.26} _{85.81}	91.61 ^{92.89} _{90.74}	81.04 ^{82.68} _{79.58}	79.6 ^{81.27} _{78.07}	82.96 ^{84.53} _{81.5}	91.34 ^{92.68} _{89.94}
KLM (Hendrycks et al., 2022)	78.8 ^{80.32} _{77.58}	82.76 ^{84.24} _{81.63}	76.9 ^{78.56} _{75.38}	76.03 ^{77.65} _{74.52}	80.69 ^{82.35} _{79.14}	88.41 ^{90.05} _{86.74}
VIM (Wang et al., 2022)	88.51 ^{89.98} _{87.42}	93.14 ^{94.53} _{92.25}	74.83 ^{76.5} _{73.17}	82.11 ^{83.71} _{80.69}	78.81 ^{80.45} _{77.2}	91.52 ^{92.9} _{90.05}
KNN (Sun et al., 2022)	90.7 ^{92.0} _{89.69}	93.1 ^{94.41} _{92.19}	80.25 ^{81.88} _{78.79}	82.32 ^{83.88} _{80.93}	81.75 ^{83.32} _{80.27}	93.47 ^{94.61} _{92.25}
DICE (Sun and Li, 2022)	77.79 ^{79.34} _{76.44}	85.41 ^{86.74} _{84.37}	79.15 ^{80.81} _{77.61}	79.84 ^{81.46} _{78.33}	81.97 ^{83.53} _{80.5}	91.19 ^{92.47} _{89.84}
RankFeat (Song et al., 2022)	76.33 ^{78.03} _{74.76}	70.15 ^{71.89} _{68.39}	62.22 ^{64.13} _{60.33}	67.74 ^{69.63} _{65.9}	58.57 ^{61.51} _{57.0}	38.97 ^{41.69} _{37.09}
ASH (Djurisic et al., 2022)	74.11 ^{75.61} _{72.71}	78.36 ^{79.83} _{77.02}	78.39 ^{80.03} _{76.89}	79.7 ^{81.31} _{78.23}	82.12 ^{83.63} _{80.72}	94.23 ^{95.25} _{93.11}
SHE (Zhang et al., 2023)	80.84 ^{82.3} _{79.64}	86.55 ^{87.88} _{85.55}	78.72 ^{80.37} _{77.18}	77.35 ^{78.98} _{75.8}	80.46 ^{82.01} _{79.0}	90.48 ^{91.71} _{89.17}

Table 1: Conformal AUROC metrics for several baselines from the OpenOOD benchmark. Each cell reports the classical AUROC as the main value, with the permissive AUROC⁺ as superscript and the conservative AUROC⁻ as subscript, i.e. AUROC^{AUROC⁺}_{AUROC⁻}. AUROC⁻ is obtained with the Monte Carlo method and AUROC⁺ with the DKWM method. All the conformal metrics are computed with $\delta = 0.01$. Top methods are **bolded**. When the intervals overlap, so that methods cannot be distinguished with the corresponding level of confidence, several are bolded.

more than 1 percent. On the one hand, this is significant, especially for such benchmarks where the State-of-the-art often holds by a fraction of a percentage. On the other hand, the correction is not *so* severe, and the best baselines still get very good AUROC despite the correction. In other words, the correction is large enough to manifest its importance but low enough to still be useable in practice: **it costs only roughly 1 or 2 percent in AUROC to be 99% sure that the FPR involved in the AUROC calculation is not overestimated.** *For benchmarking concerns*, it shows that **the top methods cannot be distinguished from one another** with the chosen level of confidence, because of the overlap of the interval built on AUROC⁻ and AUROC⁺. This set of methods is bolded in Table 1.

4.4.2. ADBENCH

We perform the same procedure as OpenOOD with ADBench (Han et al., 2022), which gathers many task-agnostic OOD baselines – considered Anomaly Detection (AD), hence the benchmark’s name. We conduct the experiments with "unsupervised AD" baselines, i.e. baselines that do not leverage labeled anomalies. We apply the correction with $\delta = 0.05$ and summarize the results in Figure 2. The complete results are deferred to Appendix F.

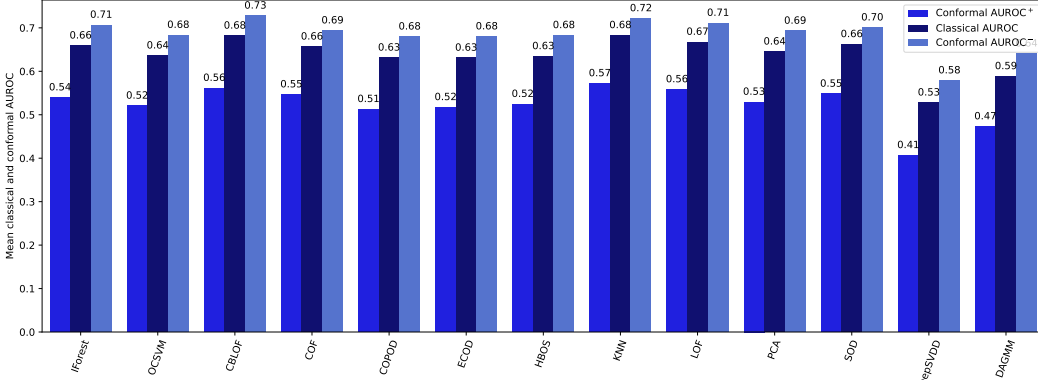


Figure 2: Results for ADBench benchmark. Mean AUROC, AUROC⁻ and AUROC⁺ over different datasets for each AD method.

Figure 2 shows a bar plot with mean classical AUROC, conformal AUROC⁻ and AUROC⁺ over all the datasets of the benchmark for all methods. The variability and magnitude of the correction are higher than for OpenOOD since the number of points in the test set changes depending on the dataset and is generally way lower. It results in larger fluctuation intervals, even for $\delta = 0.05$, which does not allow one baseline to be considered as better than the others with a probability of 0.90 (since $\delta = 0.05$). This observation is important because **it illustrates the brittleness of the conclusions that can be drawn from AD benchmarks and supports the increasingly commonly accepted fact that no method is provably better than others in AD** – one of the key conclusions of ADBench’s paper itself (Han et al., 2022). Averaged over datasets, the correction affects all baselines similarly, so the *ranking* of methods is essentially preserved even though individual comparisons are no longer significant.

5. CP Nonconformity Scores Augmented by OOD Scores

Sections 4–4.4 developed the CP→OOD direction of our bridge. We now develop the OOD→CP direction: since CP turns any score into a prediction set but tighter sets require more informative scores, the rich family of OOD scores is a natural, so-far untapped, source of nonconformity scores.

While the coverage guarantee of CP holds for any score, more informative scores lead to tighter prediction sets. This motivates considering OOD scores within CP, especially since

	LAC			APS			RAPS		
α	0.005	0.01	0.05	0.005	0.01	0.05	0.005	0.01	0.05
Cifar10									
Gram	9.57	8.34	1.89	9.60 ± 0.10	1.93 ± 0.06	8.66 ± 0.13	9.56 ± 0.13	8.7 ± 0.16	1.89 ± 0.03
ReAct	3.75	1.98	1.03	4.47 ± 0.16	1.97 ± 0.09	3.62 ± 0.17	4.46 ± 0.15	3.67 ± 0.19	2.02 ± 0.09
ODIN	7.15	5.82	1.14	7.42 ± 0.17	1.53 ± 0.06	5.14 ± 0.08	7.45 ± 0.16	5.1 ± 0.10	1.57 ± 0.08
KNN	2.57	1.48	1.01	3.62 ± 0.15	1.09 ± 0.03	2.71 ± 0.11	3.69 ± 0.11	2.77 ± 0.09	1.08 ± 0.02
Mahalanobis	1.85	<u>1.47</u>	1.04	1.89 ± 0.07	1.04 ± 0.01	1.49 ± 0.04	1.92 ± 0.05	1.49 ± 0.05	1.04 ± 0.01
CP (Softmax)	<u>2.44</u>	1.73	<u>1.03</u>	3.92 ± 0.26	1.1 ± 0.01	<u>2.16 ± 0.13</u>	3.81 ± 0.24	<u>2.17 ± 0.11</u>	1.09 ± 0.01
Cifar100									
ReAct	52.41	29.77	10.06	53.02 ± 0.27	32.02 ± 0.11	10.45 ± 0.15	53.12 ± 0.26	32.12 ± 0.13	10.43 ± 0.15
ODIN	66.54	45.25	16.25	65.46 ± 0.3	45.56 ± 0.14	17.49 ± 0.13	65.61 ± 0.25	45.51 ± 0.27	17.6 ± 0.14
KNN	41.64	27.74	8.62	<u>39.45 ± 0.35</u>	<u>29.81 ± 0.24</u>	<u>9.80 ± 0.11</u>	<u>39.63 ± 0.21</u>	<u>29.74 ± 0.29</u>	<u>9.81 ± 0.10</u>
Mahalanobis	31.29	24.76	<u>7.57</u>	31.07 ± 0.07	24.77 ± 0.20	8.47 ± 0.29	31.15 ± 0.07	24.82 ± 0.19	8.48 ± 0.21
CP (Softmax)	<u>31.96</u>	<u>27.21</u>	5.73	46.55 ± 1.47	36.82 ± 0.39	17.59 ± 0.41	45.64 ± 1.19	36.83 ± 0.79	17.12 ± 0.74

Table 2: Efficiency (mean $\pm$ std. dev. for APS and RAPS) of the prediction sets for different scores for CP classification on CIFAR-10 and CIFAR-100. Efficiency is the average prediction-set size, so *lower is better*. The best is bolded, the second is underlined.

some standard CP scores are already closely related to OOD scores such as maximum softmax. We therefore evaluate several OOD scores within the CP pipeline. We consider two ResNet18 trained on CIFAR-10 and CIFAR-100 and build conformal prediction sets following the procedure described in section 2. To build these prediction sets, we use scores based on ReAct (Sun et al., 2021), Gram (Sastry and Oore, 2020), KNN (Sun et al., 2022), Mahalanobis (Lee et al., 2018), and ODIN (Liang et al., 2018). Note that we had to adapt those scores to make them class-dependent since the score used in CP is defined as $s_{cp}(\mathbf{x}, y)$. We did so following a procedure that we describe in detail in Appendix D. We adopt throughout the OOD convention that a higher score means more anomalous, so that a candidate label y is *more* conforming when $s(\mathbf{x}, y)$ is *lower*. Given the class-dependent OOD score $s(\mathbf{x}, y_i)$, we therefore construct softmax-like scores $\hat{s}(\mathbf{x}, y_i) = \exp(-s(\mathbf{x}, y_i)) / \sum_j \exp(-s(\mathbf{x}, y_j))$ and use $\hat{s}$ as the input to the LAC/APS/RAPS pipelines exactly as the softmax output is used for the standard CP scores. The minus sign is essential: for distance-based scores such as KNN or Mahalanobis, the true class has the *smallest* distance, and $\exp(-s)$ ensures it receives the largest softmax-like mass (and hence the smallest nonconformity), whereas $\exp(s)$ would invert this ordering.

For each defined score, we perform the calibration step on $n_{cal} = 2000$ points following the classical Least-Ambiguous set classifiers (LAC) procedure (Sadinle et al., 2019), and the more recent Adaptive Prediction Set (APS) (Romano et al., 2020) and Regularized Adaptive Prediction Set (RAPS) (Angelopoulos et al., 2020) methods. For all methods, we construct the prediction sets for each of the remaining $n_{val} - n_{cal} = 8000$ points, and for risk levels $\alpha \in \{0.005, 0.01, 0.05\}$ (i.e. target coverages $1 - \alpha \in \{0.995, 0.99, 0.95\}$). We assess the mean efficiency of the prediction sets for each score, including LAC, APS, and RAPS based on softmax, as classically done in CP in Table 2. Since APS and RAPS involve sampling a uniform random variable, we report the mean and the standard deviation of the mean efficiency for 10 evaluations. Because the coverage guarantee of split CP equation (2) holds for *any* score, including the OOD-derived ones, all variants are valid by construction; we

verified that the empirical coverage matched the target $1 - \alpha$ for every score, method and risk level, so efficiency is the only quantity that differentiates them.

Table 2 shows that OOD scores are not uniformly effective for CP. For instance, Gram performs poorly, whereas KNN and Mahalanobis outperform classical CP scores in several settings. Gram is omitted from the CIFAR-100 block of Table 2 because of its poor performance, consistent with it being the weakest score on CIFAR-10. These results suggest that some OOD scores are promising candidates as nonconformity scores, even though they should not be viewed as universal replacements.

We deliberately keep this study focused on image classification and a fixed set of OOD baselines, and view it as an illustration rather than an exhaustive benchmark. Nonetheless, the pattern is informative about where gains should be expected more broadly. The scores that help, KNN and Mahalanobis, exploit class-conditional geometry in feature space, so we expect similar benefits wherever such geometry is informative but the base softmax is not already well separated, e.g. on other backbones, on tabular data, or, through their regression counterparts (nearest-neighbor and residual-based distances), in conformal regression. Conversely, gains should shrink when the classifier’s softmax already separates the classes well, in which case softmax-derived OOD scores add little over standard CP scores. A systematic study across modalities, tasks and a larger score library is a natural next step that our framing directly opens.

6. Conclusion & Discussion

In this paper, we have studied several connections between Out-of-Distribution (OOD) detection and Conformal Prediction (CP). By reframing OOD evaluation within the statistical framework provided by CP, we introduced conformal AUROC⁺ / AUROC[−] and conformal FPR@TPR95⁺ / FPR@TPR95[−], which account for the variability induced by finite validation datasets and improve the interpretation of benchmark results on OpenOOD and ADBench. In the other direction (OOD→CP), we showed that some OOD scores, notably KNN and Mahalanobis, can improve the efficiency of conformal prediction sets. Our empirical study of OOD-derived nonconformity scores is limited to image classification and to a fixed set of OOD baselines. **Taken together, these results suggest that OOD detection and CP may benefit from each other, and that this relationship deserves further investigation.**

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Appendix A. Appendix: Definition of h^-

Let $0 \leq b_1 \leq b_2 \leq \dots \leq b_n \leq 1$ such that

$$\mathbb{P}[U_{(1)} \leq b_1, \dots, U_{(n)} \leq b_n] \geq 1 - \delta.$$

The correction $h^- : \mathbb{R} \mapsto \mathbb{R}$ is defined as

$$h^-(t) = b_{\lfloor (n_{val}+1)(1-t) \rfloor}, t \in [0, 1],$$

Appendix B. Appendix: Proof of Proposition 1

In this section, we prove Proposition 1, i.e. we design calibration-conditional p-values $\hat{u}^{cc+}$ that satisfy the following property:

$$\mathbb{P}\left(\mathbb{P}(\hat{u}^{cc+}(\mathbf{x}) \leq t \mid \mathcal{D}_{id}^{val}) \geq t, \forall t \in (0, 1)\right) \geq 1 - \delta, \quad (11)$$

which allows to obtain $\widehat{\text{FPR}}^+$ that satisfies

$$\mathbb{P}\left[\text{FPR}(\tau) \geq \widehat{\text{FPR}}^+(\tau), \forall \tau \in \mathbb{R}\right] \geq 1 - \delta. \quad (12)$$

Recall the statement of Proposition 1: given $0 \leq b_1 \leq b_2 \leq \dots \leq b_n \leq 1$ such that $\mathbb{P}[U_{(1)} \leq b_1, \dots, U_{(n)} \leq b_n] \geq 1 - \delta$, and $h^+ : \mathbb{R} \mapsto \mathbb{R}$ defined by $h^+(t) = 1 - b_{\lfloor (n+1)(1-t) \rfloor}$ for $t \in [0, 1]$, and supposing that the distribution of $s(\mathbf{x})$ is continuous, $\hat{u}^{cc+}$ defined as $\hat{u}^{cc+} = h^+ \circ \hat{u}^{\text{marg}}$ satisfies equation (11) and $\widehat{\text{FPR}}^+$ defined as $\widehat{\text{FPR}}^+(\tau) = h^+ \circ \widehat{\text{FPR}}(\tau)$ satisfies equation (12).

Proof Let $U_1, \dots, U_n$ be i.i.d. uniform random variables on $[0, 1]$, with order statistics $U_{(1)} \leq \dots \leq U_{(n)}$. Let $0 \leq b_1 \leq b_2 \leq \dots \leq b_n \leq 1$ such that

$$\mathbb{P}[U_{(1)} \leq b_1, \dots, U_{(n)} \leq b_n] \geq 1 - \delta. \quad (13)$$

We can adapt the proof of Theorem 4 of Bates et al. (2023) to show that h^+ satisfies equation (12). Let $x \sim \mathcal{P}_{id}$, $S_i = s(x_i)$ for $i \in \{1, \dots, n\}$ with $F^-(t) = \mathbb{P}[s(x) < t \mid \mathcal{D}_{id}^{val}]$ and $S_{(1)} \leq S_{(2)} \leq \dots \leq S_{(n)}$ be the order statistics. Since we condition on $\mathcal{D}_{id}^{val}$, S_i are independent. Then, $\forall (z_1, \dots, z_n) \in [0, 1]^n$, since $s(\mathbf{x})$ is continuous in distribution, we have

$$\mathbb{P}[F^-(S_{(1)}) \leq z_1, \dots, F^-(S_{(n)}) \leq z_n] = \mathbb{P}[U_{(1)} \leq z_1, \dots, U_{(n)} \leq z_n], \quad (14)$$

and therefore,

$$\mathbb{P}[1 - F^-(S_{(n)}) \leq z_1, \dots, 1 - F^-(S_{(1)}) \leq z_n] = \mathbb{P}[U_{(1)} \leq z_1, \dots, U_{(n)} \leq z_n], \quad (15)$$

Let $\mathcal{E}_n$ denote the event on which $1 - F^-(S_{(i)}) \leq b_{n+1-i} \iff F^-(S_{(i)}) \geq 1 - b_{n+1-i}$ for all $i = 1, \dots, n$. Then,

$$\mathbb{P}[\mathcal{E}_n] \geq 1 - \delta.$$

Now we prove the following claim, which directly yields the theorem.

$$\mathbb{P}[\hat{u}^{cc+}(x) \leq t \mid \mathcal{D}] \geq t, \quad \forall t \in [0, 1], \text{ if } \mathcal{E}_n \text{ occurs.} \quad (16)$$

Note that the image of $\hat{u}^{\text{cc}+}$ is $\{0, 1 - b_n, \dots, 1 - b_1\}$, it remains to prove equation (16) with $t \in \{0, 1 - b_n, \dots, 1 - b_1\}$. When $t = 0$, it clearly holds. When $t = 1 - b_i$,

$$\hat{u}^{\text{cc}+}(x) \leq 1 - b_{n+1-i} \iff \hat{u}^{\text{marg}}(x) \leq \frac{i}{n+1} \iff s(x) < S_{(i)}.$$

Thus,

$$\mathbb{P}[\hat{u}^{\text{cc}+}(x) \leq 1 - b_{n+1-i} | \mathcal{D}] = \mathbb{P}[s(x) < S_{(i)} | \mathcal{D}] = F^-(S_{(i)}).$$

Hence, when $\mathcal{E}_n$ occurs, i.e. $F^-(S_{(i)}) \geq 1 - b_{n+1-i}$, we have that $\mathbb{P}[\hat{u}^{\text{cc}+}(X_{n+1}) \leq 1 - b_{n+1-i} | \mathcal{D}] \geq 1 - b_{n+1-i}$ which concludes the proof. $\blacksquare$

Appendix C. Appendix: Simes and Monte Carlo corrections

In our work we use two of the corrections proposed by Bates et al. (2023), Simes and Monte Carlo Correction. In this section, we introduce these corrections for the sake of completeness, as well as two other corrections that we do not use for reasons to be detailed.

Simes Correction Generally, we are interested in small p-values and the Simes correction focuses on those, that is by adding a smaller correction to the smaller p-values than the larger ones.

$$b_{n+1-i}^s = 1 - \delta^{2/n} \left(\frac{i \cdots (i - n/2 + 1)}{n \cdots (n - n/2 + 1)} \right)^{2/n}, \quad i = 1, \dots, n \quad (17)$$

DKWM The former approach may be compared to the classical uniform concentration DKWM result, where the b are defined as

$$b_i^d = \min\{(i/n) + \sqrt{\log(2/\delta)/2n}, 1\}; \quad (18)$$

However, DKWM tends to provide much larger bounds than Simes.

Asymptotic Correction The previous approach brought finite sample guarantees but at the cost of a large correction. In order to produce a tighter bound, for a more powerful test, we look into a correction that is correct asymptotically.

$$c_n(\delta) := \left(\sqrt{2 \log \log n} \right)^{-1} (-\log[-\log(1 - \delta)]) + 2 \log \log n + (1/2) \log \log \log n - (1/2) \log \pi. \quad (19)$$

$$b_i^a = \min \left\{ \frac{i}{n} + c_n(\delta) \frac{\sqrt{i(n-i)}}{n\sqrt{n}}, 1 \right\}, \quad i = 1, \dots, n \quad (20)$$

This bound is quite similar to Simes for small values, but quite tighter for the remaining ones.

Monte Carlo Correction The Monte Carlo Correction offers advantages of both the Simes and Asymptotic methods. It provides a finite-sample guarantee, mimics Simes for small p-values and remains closer to the asymptotic correction for larger ones.

$$h^{\text{m},\hat{\delta}}(t) = \min \left\{ h^s(t), h^{\text{a},\hat{\delta}}(t) \right\}, \quad t \in [0, 1]. \quad (21)$$

Appendix D. Appendix: Designing class-dependent OOD scores for CP

Let's consider a classification task with a classifier f trained to fit a dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, where $\mathbf{x}_i \in \mathcal{X}$ and $y_i \in \{1, \dots, C\}$ for all $i \in \{1, \dots, n\}$. In OOD, the score function $s : \mathcal{X} \rightarrow \mathbb{R}$, whereas in CP, the non-conformity score $s_{cp} : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$. Hence, in order to construct a non-conformity score out of s , we have to make it class-dependent. In this section, we describe how to construct class-dependent OOD scores out of classical OOD scores for appropriate usage in CP.

D.1. ReAct

ReAct method (Sun et al., 2021) gets the quantiles of f 's penultimate layer's activation values and then clips the activation values for a new input data point. The output softmax are then used for OOD scoring. Therefore, making the score class-dependent is straightforward: one only has to get the class softmax.

D.2. ODIN

The idea of ODIN (Liang et al., 2018) is also to tweak the network so that the softmax becomes more informative for OOD detection. Similarly to ReAct, one only has to get each class's softmax to make the score class-dependent.

D.3. KNN

For each $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ from the training set, consider $\mathbf{H} = \{h(\mathbf{x}_1), \dots, h(\mathbf{x}_n)\}$ where $h : \mathcal{X} \rightarrow \mathbb{R}^p$ is defined such that $h(\mathbf{x}_i)$ is the activation vector of $\mathbf{x}_i$ of f 's penultimate layer. Let $N_{\mathbf{H}} : \mathbb{R}^p \rightarrow \mathbb{R}^p$ be the nearest neighbor map such that $N(\mathbf{h})$ is the nearest neighbor of $\mathbf{h}$ among $\mathbf{H}$. KNN (Sun et al., 2022) builds the score s as

$$s(\mathbf{x}_{n+1}) = \|h(\mathbf{x}_{n+1}) - N_{\mathbf{H}}(h(\mathbf{x}_{n+1}))\|.$$

To make this score class-dependent, one can build C maps $\{N_{\mathbf{H}_1}, \dots, N_{\mathbf{H}_C}\}$ where $\mathbf{H}_k = \{h(\mathbf{x}_i) | f(\mathbf{x}_i) = k\}$ and then define a new score

$$s(\mathbf{x}_{n+1}, y) = \|h(\mathbf{x}_{n+1}) - N_{\mathbf{H}_y}(h(\mathbf{x}_{n+1}))\|$$

D.4. Mahalanobis

Let us consider the map h as in KNN. For each $k \in \{1, \dots, C\}$, Mahalanobis distance method (Lee et al., 2018) computes Σ_k and μ_k , which are the empirical covariance matrix and mean vectors of each set of points $\{h(\mathbf{x}_i)\}_{i|f(\mathbf{x}_i)=k}$. Then, the score s is computed as:

$$s(\mathbf{x}_{n+1}) = \sqrt{(\mathbf{x}_{n+1} - \mu_{f(\mathbf{x}_{n+1})})^T \Sigma^{-1} (\mathbf{x}_{n+1} - \mu_{f(\mathbf{x}_{n+1})})},$$

where $\Sigma = \frac{1}{C} \sum_{k \in \{1, \dots, C\}} \Sigma_k$. To make the score class-dependent, one simply has to define

$$s(\mathbf{x}_{n+1}, y) = \sqrt{(\mathbf{x}_{n+1} - \mu_y)^T \Sigma_y^{-1} (\mathbf{x}_{n+1} - \mu_y)}.$$

D.5. Gram

Let f be a classifier of depth L . Gram method (Sastry and Oore, 2020) builds a statistic $\delta : \mathcal{X} \rightarrow \mathbb{R}^L$ that outputs the channel-wise correlation of the activation maps for each layer. First, $\{\delta(\mathbf{x}_1), \dots, \delta(\mathbf{x}_n)\}$ are computed. Then, a multi-dimensional statistic $\{d_{l,k}\}_{l \in \{1, \dots, L\}, k \in \{1, \dots, C\}}$ is computed for each layer after a class-wise aggregation.

For a new test point $\mathbf{x}_{n+1}$, $\delta(\mathbf{x}_{n+1})$ is computed, along with $f(\mathbf{x}_{n+1})$. The score is built out of a weighted mean of the layer-wise deviation:

$$s(\mathbf{x}_{n+1}) = \sum_{l \in \{1, \dots, L\}} w_l |\delta(\mathbf{x}_{n+1})_l - d_{l, f(\mathbf{x}_{n+1})}|,$$

where $\{w_l\}_{l \in \{1, \dots, L\}}$ are some normalization weights computed with the training data. It is quite straightforward to make this OOD score class-dependent by defining

$$s(\mathbf{x}_{n+1}, y) = \sum_{l \in \{1, \dots, L\}} w_l |\delta(\mathbf{x}_{n+1})_l - d_{l,y}|.$$

Appendix E. Appendix: Complementary results on OpenOOD benchmark

In this section, we present the full results of benchmarks on OpenOOD. The results displayed are AUROC with $\delta = 0.05$ in Table 3, FPR@TPR95 with $\delta = 0.05$ in Table 5 and FPR@TPR95 with $\delta = 0.01$ in Table 4. In every table we report the classical (marginal) metric next to its conformal counterpart, so that the magnitude of the correction can be read directly. Two trends complement the discussion in Section 4.4.1: (i) the conformal AUROC is systematically *lower* than the classical one (conservative), whereas the conformal FPR@TPR95 is systematically *higher* (also conservative, since a larger FPR is the pessimistic direction); and (ii) relaxing the risk from $\delta = 0.01$ to $\delta = 0.05$ shrinks the correction, as expected, since a larger tolerated miscalibration probability yields a tighter bound. The relative ordering of the strongest baselines is largely preserved, confirming that the correction is informative without overturning the benchmark conclusions.

	CIFAR-10				CIFAR-100				ImageNet-200			
OOD type	Near		Far		Near		Far		Near		Far	
	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.
OpenMax (Bendale and Boulton, 2015)	87.2	86.18	89.53	88.52	76.66	75.26	79.12	77.81	80.4	79.2	90.41	89.49
MSP (Hendrycks and Gimpel, 2016)	87.68	86.76	91.0	90.17	80.42	79.2	77.58	76.29	83.3	82.2	90.2	89.36
TempScale (Guo et al., 2017)	87.65	86.75	91.27	90.48	80.98	79.78	78.51	77.24	83.66	82.58	90.91	90.11
ODIN (Liang et al., 2018)	80.25	79.26	87.21	86.43	79.8	78.57	79.44	78.2	80.32	79.19	91.89	91.17
MDS (Lee et al., 2018)	86.72	85.71	90.2	89.29	58.79	57.2	70.06	68.63	62.51	60.96	74.94	73.6
MDSEns (Lee et al., 2018)	60.46	59.01	74.07	72.96	45.98	44.34	66.03	64.72	54.58	52.99	70.08	68.76
Gram (Sastry and Oore, 2020)	52.63	51.04	69.74	68.41	50.69	49.06	73.97	72.87	68.36	67.0	70.94	69.69
EBO (Liu et al., 2020)	86.93	86.08	91.74	91.05	80.84	79.63	79.71	78.47	82.57	81.47	91.12	90.33
GradNorm (Huang et al., 2021)	53.77	52.26	58.55	57.09	69.73	68.41	68.82	67.48	73.33	72.12	85.29	84.45
ReAct (Sun et al., 2021)	86.47	85.6	91.02	90.28	80.7	79.5	79.84	78.6	80.48	79.35	93.1	92.4
MLS (Hendrycks et al., 2022)	86.86	86.0	91.61	90.9	81.04	79.84	79.6	78.35	82.96	81.88	91.34	90.56
KLM (Hendrycks et al., 2022)	78.8	77.8	82.76	81.83	76.9	75.65	76.03	74.8	80.69	79.54	88.41	87.44
VIM (Wang et al., 2022)	88.51	87.62	93.14	92.41	74.83	73.47	82.11	80.95	78.81	77.57	91.52	90.7
KNN (Sun et al., 2022)	90.7	89.87	93.1	92.35	80.25	79.05	82.32	81.19	81.75	80.63	93.47	92.83
DICE (Sun and Li, 2022)	77.79	76.68	85.41	84.56	79.15	77.89	79.84	78.61	81.97	80.86	91.19	90.43
RankFeat (Song et al., 2022)	76.33	75.05	70.15	68.71	62.22	60.67	67.74	66.24	58.57	57.06	38.97	37.43
ASH (Djurisic et al., 2022)	74.11	72.96	78.36	77.27	78.39	77.16	79.7	78.5	82.12	81.07	94.23	93.66
SHE (Zhang et al., 2023)	80.84	79.86	86.55	85.73	78.72	77.46	77.35	76.08	80.46	79.34	90.48	89.72

Table 3: Classical AUROC (marg.) vs Conformal AUROC (conf.) obtained with the Monte Carlo method and $\delta = 0.05$ for several baselines from OpenOOD benchmark.

	CIFAR-10				CIFAR-100				ImageNet-200			
OOD type	Near		Far		Near		Far		Near		Far	
	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.
OpenMax (Bendale and Boulton, 2015)	46.77	48.98	29.48	31.48	55.57	57.8	54.77	57.0	63.32	65.75	32.29	35.35
MSP (Hendrycks and Gimpel, 2016)	53.57	55.8	31.44	33.45	54.73	56.96	59.08	61.31	55.25	57.69	35.44	38.29
TempScale (Guo et al., 2017)	56.85	59.08	33.36	35.38	54.77	56.99	58.24	60.47	55.03	57.5	34.11	37.06
ODIN (Liang et al., 2018)	84.55	86.78	60.9	62.97	58.44	60.67	57.75	59.98	66.38	68.8	33.66	36.75
MDS (Lee et al., 2018)	46.22	48.44	30.3	32.3	82.75	84.98	70.46	72.68	79.34	81.52	61.26	63.81
MDSEns (Lee et al., 2018)	92.06	94.29	61.09	62.87	95.84	98.07	66.97	68.85	91.69	93.8	80.43	82.89
Gram (Sastry and Oore, 2020)	93.52	95.75	69.29	71.48	92.48	94.71	63.1	65.2	85.43	87.63	84.95	87.44
EBO (Liu et al., 2020)	67.54	69.77	40.55	42.58	55.49	57.72	56.41	58.64	59.46	61.93	34.0	37.07
GradNorm (Huang et al., 2021)	95.37	97.6	89.34	91.52	86.13	88.36	82.79	85.02	83.07	85.33	66.78	69.67
ReAct (Sun et al., 2021)	71.56	73.78	42.43	44.52	56.74	58.97	56.32	58.55	65.37	67.8	27.21	30.28
MLS (Hendrycks et al., 2022)	67.54	69.77	40.53	42.56	55.48	57.71	56.53	58.76	58.94	61.44	33.59	36.68
KLM (Hendrycks et al., 2022)	86.41	88.63	76.42	78.65	79.52	81.75	70.16	72.39	69.42	71.91	39.57	42.56
VIM (Wang et al., 2022)	48.07	50.29	25.77	27.65	62.96	65.19	49.72	51.95	59.91	62.32	26.86	29.81
KNN (Sun et al., 2022)	34.54	36.65	23.88	25.77	61.32	63.54	54.04	56.27	60.42	62.9	26.49	29.66
DICE (Sun and Li, 2022)	80.15	82.38	53.93	56.06	58.1	60.33	55.95	58.17	60.98	63.46	35.93	39.04
RankFeat (Song et al., 2022)	67.38	69.61	68.24	70.47	79.94	82.17	68.89	71.11	92.02	93.91	98.48	99.58
ASH (Djurisic et al., 2022)	89.03	91.26	76.66	78.89	66.14	68.37	62.67	64.89	65.95	68.44	26.26	29.46
SHE (Zhang et al., 2023)	84.49	86.72	63.26	65.41	59.32	61.54	62.74	64.97	65.92	68.31	41.5	44.62

Table 4: Classical FPR@TPR95 (marg.) vs Conformal FPR@TPR95 (conf.) obtained with the Monte Carlo method and $\delta = 0.01$ for several baselines from OpenOOD benchmark.

Appendix F. Appendix: Full results for ADBench

In this section, we present the full results of the ADBench benchmark, broken down by dataset (rows) and method (columns). Table 6 displays classical AUROC, Table 7 displays conformal AUROC, and Table 8 displays the difference between the two (AUROC correction), all with $\delta = 0.05$. The per-dataset breakdown makes explicit the point raised in Section 4.4.2:

	CIFAR-10				CIFAR-100				ImageNet-200			
OOD type	Near		Far		Near		Far		Near		Far	
	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.	marg.	conf.
OpenMax (Bendale and Boulton, 2015)	46.77	48.58	29.48	31.11	55.57	57.39	54.77	56.59	63.32	65.14	32.29	33.98
MSP (Hendrycks and Gimpel, 2016)	53.57	55.39	31.44	33.08	54.73	56.55	59.08	60.9	55.25	57.06	35.44	37.16
TempScale (Guo et al., 2017)	56.85	58.67	33.36	35.01	54.77	56.59	58.24	60.06	55.03	56.85	34.11	35.8
ODIN (Liang et al., 2018)	84.55	86.37	60.9	62.59	58.44	60.26	57.75	59.57	66.38	68.2	33.66	35.34
MDS (Lee et al., 2018)	46.22	48.03	30.3	31.94	82.75	84.57	70.46	72.28	79.34	81.16	61.26	63.08
MDSEns (Lee et al., 2018)	92.06	93.88	61.09	62.54	95.84	97.66	66.97	68.5	91.69	93.51	80.43	82.25
Gram (Sastry and Oore, 2020)	93.52	95.34	69.29	71.08	92.48	94.3	63.1	64.81	85.43	87.25	84.95	86.77
EBO (Liu et al., 2020)	67.54	69.36	40.55	42.21	55.49	57.31	56.41	58.23	59.46	61.28	34.0	35.7
GradNorm (Huang et al., 2021)	95.37	97.19	89.34	91.16	86.13	87.95	82.79	84.61	83.07	84.89	66.78	68.6
ReAct (Sun et al., 2021)	71.56	73.38	42.43	44.14	56.74	58.56	56.32	58.14	65.37	67.19	27.21	28.81
MLS (Hendrycks et al., 2022)	67.54	69.36	40.53	42.19	55.48	57.3	56.53	58.35	58.94	60.76	33.59	35.28
KLM (Hendrycks et al., 2022)	86.41	88.23	76.42	78.24	79.52	81.34	70.16	71.98	69.42	71.24	39.57	41.3
VIM (Wang et al., 2022)	48.07	49.88	25.77	27.3	62.96	64.78	49.72	51.54	59.91	61.72	26.86	28.46
KNN (Sun et al., 2022)	34.54	36.27	23.88	25.42	61.32	63.14	54.04	55.86	60.42	62.23	26.49	28.09
DICE (Sun and Li, 2022)	80.15	81.97	53.93	55.67	58.1	59.92	55.95	57.77	60.98	62.8	35.93	37.66
RankFeat (Song et al., 2022)	67.38	69.2	68.24	70.06	79.94	81.76	68.89	70.71	92.02	93.84	98.48	99.55
ASH (Djurisic et al., 2022)	89.03	90.85	76.66	78.48	66.14	67.96	62.67	64.49	65.95	67.77	26.26	27.85
SHE (Zhang et al., 2023)	84.49	86.31	63.26	65.02	59.32	61.14	62.74	64.56	65.92	67.74	41.5	43.27

Table 5: Classical FPR@TPR95 (marg.) vs Conformal FPR@TPR95 (conf.) obtained with the Monte Carlo method and $\delta = 0.05$ for several baselines from OpenOOD benchmark.

the magnitude of the correction (Table 8) is driven by the number of test points available for each dataset, so datasets with few samples (e.g. *musk*, *satimage-2*) incur much larger corrections than large ones (e.g. *cover*, *donors*). Because these corrections are often of the same order as the gap between competing methods on a given dataset, they reinforce the conclusion that no single unsupervised AD method can be declared uniformly superior at the chosen confidence level.

	IForest	OCSVM	CBLOF	COF	COPOD	ECOD	HBOS	KNN	LOF	PCA	SOD	DeepSVDD	DAGMM
cover	0.87	0.93	0.89	0.77	0.89	0.92	0.80	0.86	0.85	0.94	0.74	0.46	0.90
donors	0.78	0.72	0.62	0.71	0.82	0.89	0.78	0.82	0.59	0.83	0.56	0.36	0.71
fault	0.57	0.48	0.64	0.62	0.44	0.45	0.51	0.73	0.59	0.46	0.68	0.52	0.46
fraud	0.90	0.91	0.88	0.96	0.88	0.89	0.90	0.93	0.96	0.90	0.95	0.73	0.90
glass	0.77	0.35	0.83	0.72	0.72	0.66	0.77	0.82	0.69	0.66	0.73	0.47	0.76
Hepatitis	0.70	0.68	0.66	0.41	0.82	0.75	0.80	0.53	0.38	0.76	0.68	0.52	0.55
Ionosphere	0.84	0.76	0.91	0.87	0.79	0.73	0.62	0.88	0.91	0.79	0.86	0.51	0.73
landsat	0.48	0.36	0.64	0.53	0.42	0.36	0.55	0.58	0.54	0.36	0.60	0.63	0.44
ALOI	0.57	0.56	0.55	0.65	0.54	0.56	0.53	0.61	0.67	0.57	0.61	0.51	0.52
letter	0.61	0.46	0.76	0.80	0.54	0.56	0.60	0.86	0.84	0.50	0.84	0.56	0.50
20news 0	0.64	0.63	0.71	0.71	0.61	0.61	0.62	0.73	0.80	0.64	0.73	0.50	0.63
20news 1	0.51	0.53	0.52	0.58	0.52	0.54	0.53	0.57	0.61	0.54	0.58	0.48	0.54
20news 2	0.50	0.51	0.47	0.53	0.50	0.52	0.51	0.51	0.54	0.51	0.50	0.49	0.53
20news 3	0.75	0.72	0.83	0.81	0.75	0.75	0.74	0.79	0.71	0.73	0.70	0.67	0.54
20news 4	0.48	0.51	0.45	0.57	0.48	0.51	0.50	0.48	0.51	0.51	0.53	0.53	0.48
20news 5	0.52	0.49	0.47	0.50	0.48	0.46	0.49	0.48	0.55	0.48	0.48	0.49	0.54
Lymphography	1.00	1.00	1.00	0.91	0.99	1.00	0.99	0.56	0.90	1.00	0.73	0.34	0.72
magic.gamma	0.73	0.61	0.75	0.67	0.68	0.64	0.71	0.82	0.69	0.67	0.75	0.60	0.59
musk	1.00	0.81	1.00	0.39	0.94	0.95	1.00	0.70	0.41	1.00	0.74	0.56	0.77
PageBlocks	0.90	0.89	0.85	0.73	0.88	0.92	0.81	0.82	0.76	0.91	0.78	0.59	0.90
pendigits	0.95	0.94	0.90	0.45	0.91	0.93	0.93	0.73	0.48	0.94	0.66	0.42	0.64
Pima	0.73	0.67	0.71	0.61	0.69	0.63	0.71	0.73	0.66	0.71	0.61	0.51	0.56
anthyroid	0.82	0.57	0.62	0.66	0.77	0.79	0.60	0.72	0.70	0.66	0.77	0.77	0.57
satellite	0.70	0.59	0.71	0.55	0.63	0.58	0.75	0.65	0.56	0.60	0.64	0.55	0.62
satimage-2	0.99	0.97	1.00	0.57	0.97	0.96	0.98	0.93	0.47	0.98	0.83	0.49	0.96
shuttle	1.00	0.97	0.83	0.52	0.99	0.99	0.99	0.70	0.57	0.99	0.70	0.49	0.98
smtp	0.86	0.72	0.70	0.69	0.70	0.78	0.56	0.84	0.58	0.83	0.40	0.72	0.71
speech	0.51	0.50	0.51	0.56	0.53	0.51	0.51	0.51	0.52	0.51	0.56	0.54	0.53
Stamps	0.91	0.84	0.68	0.54	0.93	0.88	0.91	0.69	0.51	0.91	0.73	0.56	0.89
thyroid	0.98	0.88	0.95	0.91	0.94	0.98	0.96	0.96	0.87	0.96	0.93	0.49	0.80
vertebral	0.37	0.38	0.41	0.49	0.26	0.41	0.29	0.34	0.49	0.37	0.40	0.37	0.53
vowels	0.75	0.63	0.90	0.95	0.55	0.62	0.73	0.97	0.93	0.67	0.92	0.56	0.61
Waveform	0.71	0.56	0.72	0.73	0.75	0.62	0.69	0.74	0.73	0.65	0.69	0.56	0.49
WDBC	0.99	0.99	0.99	0.96	0.99	0.97	0.99	0.92	0.89	0.99	0.92	0.62	0.77
Wilt	0.42	0.31	0.33	0.50	0.33	0.36	0.32	0.48	0.51	0.20	0.53	0.46	0.37
wine	0.80	0.73	0.26	0.44	0.89	0.77	0.91	0.45	0.38	0.84	0.46	0.60	0.62
WPBC	0.47	0.45	0.45	0.46	0.49	0.47	0.51	0.47	0.41	0.46	0.51	0.50	0.48
yeast	0.38	0.41	0.45	0.44	0.37	0.44	0.40	0.39	0.45	0.41	0.42	0.48	0.41
campaign	0.73	0.67	0.64	0.58	0.78	0.77	0.79	0.73	0.59	0.73	0.69	0.53	0.58
cardio	0.93	0.94	0.90	0.71	0.92	0.94	0.85	0.77	0.66	0.96	0.73	0.58	0.75
Cardiotocography	0.68	0.78	0.65	0.54	0.67	0.78	0.61	0.56	0.60	0.75	0.52	0.53	0.62
celeba	0.70	0.71	0.74	0.39	0.76	0.76	0.76	0.60	0.39	0.79	0.48	0.54	0.45
CIFAR10 0	0.73	0.68	0.70	0.70	0.69	0.70	0.70	0.74	0.74	0.70	0.71	0.56	0.53
CIFAR10 1	0.55	0.59	0.61	0.63	0.46	0.51	0.44	0.60	0.72	0.60	0.62	0.50	0.58
CIFAR10 2	0.56	0.58	0.58	0.61	0.56	0.57	0.54	0.60	0.65	0.58	0.59	0.58	0.51
CIFAR10 3	0.55	0.58	0.59	0.56	0.51	0.53	0.50	0.56	0.60	0.56	0.56	0.60	0.56
CIFAR10 5	0.50	0.58	0.58	0.57	0.47	0.52	0.47	0.54	0.60	0.57	0.54	0.46	0.59
CIFAR10 6	0.64	0.65	0.68	0.69	0.65	0.66	0.65	0.72	0.72	0.68	0.69	0.57	0.50
CIFAR10 7	0.54	0.59	0.56	0.57	0.52	0.55	0.50	0.54	0.60	0.57	0.56	0.62	0.61
agnews 0	0.50	0.47	0.54	0.61	0.49	0.47	0.48	0.58	0.63	0.47	0.56	0.35	0.48
agnews 1	0.58	0.54	0.58	0.71	0.51	0.54	0.55	0.62	0.74	0.55	0.61	0.37	0.56
agnews 2	0.65	0.61	0.71	0.73	0.61	0.59	0.61	0.75	0.79	0.61	0.73	0.50	0.53
agnews 3	0.54	0.55	0.57	0.70	0.51	0.53	0.51	0.62	0.70	0.55	0.61	0.50	0.51
amazon	0.56	0.54	0.58	0.58	0.57	0.54	0.56	0.59	0.56	0.54	0.58	0.45	0.51
imdb	0.50	0.45	0.50	0.49	0.50	0.45	0.48	0.48	0.49	0.46	0.50	0.52	0.42
yelp	0.61	0.59	0.64	0.68	0.60	0.57	0.59	0.68	0.66	0.59	0.66	0.50	0.55

Table 6: Full results for ADBench: classical AUROC.

EXPLORING THE LINK BETWEEN OOD DETECTION AND CONFORMAL PREDICTION

	IForest	OCSVM	CBLOF	COF	COPOD	ECOD	HBOS	KNN	LOF	PCA	SOD	DeepSVDD	DAGMM
cover	0.75	0.83	0.79	0.64	0.78	0.82	0.74	0.74	0.73	0.84	0.60	0.30	0.79
donors	0.72	0.66	0.54	0.64	0.77	0.85	0.72	0.76	0.53	0.78	0.50	0.31	0.65
fault	0.48	0.40	0.55	0.53	0.35	0.37	0.43	0.65	0.50	0.37	0.59	0.43	0.37
fraud	0.77	0.63	0.63	0.82	0.62	0.64	0.86	0.68	0.79	0.66	0.69	0.51	0.64
glass	0.65	0.31	0.73	0.60	0.60	0.53	0.64	0.72	0.58	0.54	0.63	0.38	0.66
Hepatitis	0.54	0.52	0.52	0.23	0.70	0.61	0.66	0.26	0.22	0.62	0.54	0.38	0.40
Ionosphere	0.77	0.68	0.85	0.80	0.70	0.63	0.50	0.83	0.85	0.71	0.80	0.38	0.64
landsat	0.42	0.30	0.58	0.48	0.35	0.30	0.49	0.52	0.48	0.30	0.54	0.58	0.39
ALOI	0.49	0.46	0.45	0.55	0.43	0.46	0.46	0.52	0.58	0.47	0.52	0.41	0.42
letter	0.44	0.30	0.61	0.66	0.36	0.39	0.42	0.74	0.72	0.33	0.71	0.40	0.35
20news 0	0.51	0.49	0.57	0.60	0.48	0.46	0.47	0.62	0.69	0.50	0.61	0.36	0.49
20news 1	0.36	0.39	0.37	0.44	0.37	0.39	0.37	0.44	0.47	0.39	0.44	0.33	0.39
20news 2	0.34	0.36	0.34	0.39	0.34	0.36	0.35	0.36	0.38	0.36	0.34	0.32	0.38
20news 3	0.65	0.61	0.73	0.71	0.63	0.64	0.64	0.69	0.58	0.62	0.59	0.54	0.45
20news 4	0.28	0.31	0.27	0.39	0.27	0.32	0.30	0.30	0.34	0.31	0.35	0.35	0.30
20news 5	0.34	0.32	0.29	0.30	0.30	0.31	0.32	0.29	0.35	0.32	0.28	0.31	0.35
Lymphography	0.95	0.95	0.95	0.83	0.95	0.95	0.95	0.45	0.81	0.96	0.62	0.25	0.62
magic.gamma	0.70	0.57	0.72	0.63	0.65	0.60	0.68	0.79	0.65	0.64	0.72	0.56	0.55
musk	0.57	0.65	0.22	0.21	0.83	0.85	0.58	0.53	0.22	0.32	0.59	0.42	0.64
PageBlocks	0.84	0.83	0.79	0.67	0.82	0.86	0.74	0.76	0.70	0.85	0.71	0.52	0.84
pendigits	0.87	0.86	0.82	0.33	0.82	0.85	0.85	0.60	0.35	0.86	0.53	0.29	0.52
Pima	0.63	0.57	0.62	0.51	0.59	0.54	0.62	0.64	0.55	0.61	0.52	0.40	0.45
anthyroid	0.76	0.49	0.55	0.59	0.70	0.72	0.53	0.65	0.63	0.59	0.71	0.71	0.49
satellite	0.67	0.55	0.67	0.50	0.59	0.54	0.71	0.61	0.51	0.56	0.59	0.50	0.58
satimage-2	0.51	0.71	0.45	0.41	0.74	0.80	0.77	0.79	0.34	0.70	0.68	0.31	0.84
shuttle	0.94	0.87	0.74	0.46	0.89	0.94	0.94	0.65	0.52	0.85	0.63	0.42	0.91
smtp	0.81	0.60	0.58	0.59	0.58	0.68	0.45	0.76	0.48	0.72	0.32	0.60	0.60
speech	0.31	0.31	0.31	0.37	0.34	0.32	0.31	0.31	0.33	0.32	0.35	0.34	0.34
Stamps	0.84	0.75	0.57	0.42	0.87	0.80	0.83	0.57	0.39	0.85	0.62	0.42	0.81
thyroid	0.90	0.77	0.86	0.81	0.86	0.90	0.92	0.87	0.77	0.88	0.84	0.33	0.69
vertebral	0.23	0.26	0.28	0.37	0.14	0.29	0.17	0.20	0.37	0.25	0.29	0.24	0.40
vowels	0.57	0.45	0.73	0.76	0.35	0.45	0.54	0.77	0.74	0.49	0.75	0.35	0.43
Waveform	0.56	0.42	0.59	0.58	0.60	0.47	0.53	0.59	0.59	0.50	0.54	0.39	0.34
WDBC	0.95	0.95	0.95	0.91	0.95	0.92	0.96	0.86	0.81	0.95	0.85	0.50	0.67
Wilt	0.29	0.20	0.21	0.37	0.20	0.24	0.19	0.35	0.38	0.12	0.42	0.34	0.26
wine	0.70	0.63	0.12	0.31	0.80	0.67	0.84	0.33	0.26	0.75	0.32	0.48	0.48
WPBC	0.33	0.32	0.32	0.33	0.36	0.33	0.38	0.32	0.29	0.33	0.39	0.38	0.35
yeast	0.29	0.31	0.35	0.35	0.28	0.34	0.31	0.30	0.36	0.31	0.32	0.38	0.31
campaign	0.68	0.62	0.59	0.52	0.74	0.72	0.75	0.68	0.53	0.68	0.64	0.48	0.52
cardio	0.84	0.85	0.80	0.60	0.83	0.84	0.75	0.67	0.53	0.86	0.62	0.47	0.64
Cardiotocography	0.59	0.70	0.57	0.45	0.58	0.70	0.52	0.48	0.51	0.66	0.43	0.45	0.53
celeba	0.64	0.65	0.69	0.30	0.70	0.71	0.71	0.28	0.30	0.74	0.38	0.48	0.37
CIFAR10 0	0.63	0.59	0.60	0.60	0.59	0.60	0.60	0.65	0.64	0.61	0.61	0.46	0.42
CIFAR10 1	0.43	0.48	0.50	0.52	0.35	0.39	0.33	0.49	0.62	0.49	0.51	0.39	0.48
CIFAR10 2	0.45	0.48	0.47	0.50	0.44	0.45	0.43	0.49	0.55	0.47	0.48	0.48	0.40
CIFAR10 3	0.44	0.48	0.49	0.46	0.40	0.42	0.39	0.47	0.50	0.46	0.46	0.49	0.45
CIFAR10 5	0.38	0.48	0.47	0.46	0.35	0.40	0.34	0.42	0.49	0.46	0.42	0.34	0.49
CIFAR10 6	0.54	0.55	0.58	0.59	0.54	0.55	0.54	0.61	0.62	0.58	0.59	0.47	0.39
CIFAR10 7	0.43	0.48	0.45	0.46	0.41	0.44	0.39	0.44	0.50	0.46	0.46	0.51	0.50
agnews 0	0.41	0.39	0.45	0.53	0.40	0.38	0.39	0.49	0.56	0.39	0.48	0.27	0.40
agnews 1	0.50	0.46	0.50	0.64	0.42	0.45	0.46	0.54	0.67	0.47	0.53	0.28	0.48
agnews 2	0.57	0.53	0.63	0.66	0.52	0.51	0.52	0.68	0.73	0.53	0.66	0.42	0.45
agnews 3	0.45	0.46	0.49	0.63	0.43	0.44	0.43	0.54	0.64	0.46	0.53	0.42	0.42
amazon	0.48	0.45	0.50	0.49	0.48	0.46	0.47	0.50	0.48	0.46	0.50	0.37	0.43
imdb	0.41	0.36	0.41	0.40	0.42	0.36	0.40	0.39	0.40	0.37	0.41	0.44	0.34
yelp	0.52	0.50	0.55	0.60	0.52	0.49	0.51	0.60	0.59	0.51	0.58	0.42	0.47

Table 7: Full results for ADBench: conformal AUROC.

	IForest	OCSVM	CBLOF	COF	COPOD	ECOD	HBOS	KNN	LOF	PCA	SOD	DeepSVDD	DAGMM
cover	0.12	0.10	0.11	0.13	0.11	0.10	0.07	0.12	0.12	0.10	0.14	0.15	0.11
donors	0.06	0.07	0.08	0.07	0.05	0.04	0.06	0.06	0.06	0.05	0.06	0.05	0.06
fault	0.09	0.08	0.09	0.09	0.09	0.09	0.09	0.08	0.09	0.09	0.09	0.08	0.09
fraud	0.13	0.28	0.25	0.14	0.26	0.25	0.04	0.26	0.16	0.25	0.26	0.22	0.26
glass	0.12	0.05	0.10	0.12	0.12	0.13	0.13	0.10	0.11	0.13	0.10	0.09	0.10
Hepatitis	0.16	0.15	0.14	0.18	0.12	0.14	0.13	0.27	0.16	0.14	0.14	0.14	0.14
Ionosphere	0.08	0.08	0.06	0.07	0.09	0.10	0.12	0.05	0.05	0.08	0.06	0.13	0.09
landsat	0.06	0.06	0.05	0.05	0.06	0.06	0.06	0.06	0.05	0.06	0.05	0.05	0.05
ALOI	0.07	0.10	0.10	0.09	0.10	0.10	0.06	0.09	0.08	0.10	0.09	0.10	0.10
letter	0.17	0.16	0.15	0.14	0.19	0.17	0.18	0.13	0.13	0.17	0.13	0.15	0.15
20news 0	0.14	0.14	0.13	0.11	0.14	0.15	0.14	0.11	0.11	0.14	0.12	0.13	0.14
20news 1	0.15	0.15	0.15	0.14	0.15	0.15	0.16	0.13	0.14	0.15	0.14	0.14	0.15
20news 2	0.16	0.15	0.14	0.15	0.16	0.16	0.16	0.15	0.16	0.15	0.15	0.16	0.15
20news 3	0.10	0.11	0.10	0.10	0.11	0.11	0.10	0.10	0.13	0.11	0.12	0.13	0.09
20news 4	0.20	0.20	0.17	0.18	0.21	0.19	0.20	0.18	0.16	0.20	0.17	0.18	0.18
20news 5	0.17	0.17	0.18	0.20	0.18	0.15	0.17	0.19	0.20	0.16	0.20	0.18	0.19
Lymphography	0.05	0.05	0.05	0.08	0.04	0.05	0.05	0.11	0.09	0.04	0.11	0.09	0.11
magic.gamma	0.03	0.04	0.03	0.04	0.03	0.04	0.03	0.03	0.04	0.03	0.03	0.04	0.04
musk	0.43	0.15	0.78	0.18	0.11	0.11	0.42	0.17	0.19	0.68	0.16	0.14	0.13
PageBlocks	0.06	0.06	0.06	0.06	0.06	0.06	0.07	0.06	0.06	0.06	0.07	0.06	0.05
pendigits	0.08	0.08	0.09	0.12	0.08	0.08	0.08	0.13	0.13	0.08	0.13	0.13	0.12
Pima	0.10	0.10	0.10	0.10	0.10	0.09	0.09	0.10	0.10	0.09	0.10	0.11	0.11
amthyroid	0.06	0.08	0.07	0.07	0.07	0.06	0.07	0.07	0.07	0.07	0.06	0.06	0.08
satellite	0.04	0.04	0.04	0.05	0.04	0.04	0.04	0.05	0.05	0.04	0.05	0.05	0.04
satimage-2	0.48	0.26	0.55	0.16	0.23	0.16	0.21	0.14	0.13	0.27	0.15	0.18	0.12
shuttle	0.06	0.10	0.09	0.06	0.10	0.05	0.05	0.04	0.05	0.13	0.06	0.07	0.07
smtp	0.05	0.12	0.12	0.10	0.12	0.11	0.10	0.09	0.09	0.11	0.08	0.12	0.11
speech	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.20	0.19	0.19	0.21	0.21	0.18
Stamps	0.07	0.09	0.11	0.11	0.06	0.08	0.07	0.12	0.12	0.07	0.11	0.14	0.07
thyroid	0.08	0.10	0.09	0.10	0.08	0.08	0.04	0.09	0.10	0.08	0.09	0.16	0.11
vertebral	0.14	0.12	0.13	0.12	0.12	0.12	0.12	0.14	0.13	0.12	0.11	0.12	0.13
vowels	0.19	0.18	0.17	0.20	0.20	0.17	0.20	0.20	0.19	0.18	0.17	0.21	0.19
Waveform	0.15	0.15	0.14	0.15	0.15	0.15	0.15	0.14	0.14	0.16	0.15	0.16	0.16
WDBC	0.04	0.04	0.04	0.06	0.04	0.05	0.04	0.06	0.08	0.04	0.07	0.12	0.10
Wilt	0.13	0.11	0.12	0.12	0.13	0.12	0.14	0.13	0.13	0.08	0.12	0.12	0.11
wine	0.10	0.10	0.14	0.14	0.09	0.11	0.08	0.12	0.11	0.10	0.15	0.11	0.13
WPBC	0.13	0.13	0.13	0.13	0.13	0.13	0.14	0.14	0.12	0.13	0.12	0.12	0.13
yeast	0.09	0.10	0.10	0.10	0.09	0.09	0.09	0.09	0.10	0.10	0.11	0.10	0.10
campaign	0.05	0.05	0.05	0.06	0.05	0.05	0.04	0.05	0.06	0.05	0.06	0.05	0.06
cardio	0.09	0.09	0.10	0.12	0.09	0.09	0.10	0.10	0.14	0.10	0.11	0.11	0.11
Cardiotocography	0.09	0.08	0.08	0.09	0.09	0.08	0.08	0.08	0.09	0.09	0.08	0.08	0.09
celeba	0.06	0.06	0.05	0.09	0.05	0.05	0.05	0.31	0.09	0.05	0.10	0.06	0.08
CIFAR10 0	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.11
CIFAR10 1	0.12	0.11	0.11	0.11	0.11	0.12	0.11	0.11	0.10	0.11	0.10	0.10	0.11
CIFAR10 2	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.10	0.11
CIFAR10 3	0.11	0.11	0.10	0.10	0.11	0.11	0.11	0.10	0.10	0.11	0.10	0.11	0.11
CIFAR10 5	0.12	0.10	0.11	0.11	0.12	0.12	0.12	0.12	0.11	0.11	0.12	0.12	0.10
CIFAR10 6	0.11	0.10	0.11	0.10	0.11	0.11	0.11	0.10	0.10	0.10	0.10	0.10	0.11
CIFAR10 7	0.11	0.11	0.10	0.10	0.11	0.11	0.11	0.10	0.10	0.10	0.10	0.11	0.11
agnews 0	0.09	0.09	0.08	0.08	0.09	0.09	0.09	0.08	0.08	0.09	0.08	0.09	0.09
agnews 1	0.09	0.09	0.08	0.07	0.09	0.09	0.09	0.08	0.07	0.09	0.08	0.09	0.09
agnews 2	0.08	0.08	0.08	0.07	0.08	0.09	0.08	0.07	0.06	0.08	0.07	0.08	0.08
agnews 3	0.09	0.09	0.08	0.07	0.09	0.09	0.09	0.08	0.07	0.09	0.08	0.08	0.08
amazon	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.09	0.09	0.08	0.09	0.08	0.08
imdb	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.08	0.08
yelp	0.08	0.08	0.08	0.08	0.08	0.09	0.08	0.08	0.08	0.08	0.08	0.08	0.08

Table 8: Full results for ADBench: AUROC correction (difference between conformal and classical AUROC).

